A Guide to Forecast-Validation for Evaluating Conditional Volatility Forecast Models

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Abstract

This study analyzes the relationship between high-dimensional time series data through the lens of dynamic factor models, and the predictive capacity of such models in forecasting applications. Leveraging time series data from the Federal Reserve Bank of St. Louis FRED-QD database, we first identify the state space model representing this database, and then estimate the common factors using the expectation maximization algorithm with the Kalman smoother and augment the model by specifying a sparse VAR for the remaining idiosyncratic components by the means of L_1 penalized regression. Results reveal that dynamic factor models produce reliable forecasts of macroeconomic time series while simultaneously reducing the dimension of the database. Furthermore, we find that in the case of macroeconomic variables, once the common factors are extracted from the data, accounting for the remaining latent components in the database does not provide any meaningful information.

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1 Introduction

A critical task in portfolio management is forecasting the conditional variance of financial returns. Forecasts of the conditional variance are necessary inputs for modeling risk, pricing derivatives, and hedging portfolios. This wide-spread interest in forecasting the conditional variance has led to the rise of the enormous volatility forecasting literature, starting with the seminal paper of Engle (1982). Ever since the introduction of the ARCH and GARCH class of models (Engle (1982) and Bollerslev (1986)), a great number of models has been developed, extending the baseline ARCH framework to include more sophisticated dynamics and to accommodate for the empirical regularities observed in financial markets. See Pagan (1996) and Poon and Granger (2003) for a comprehensive review.

This impressive pace in the development of volatility forecast models, however, had remained unmatched for the most part by model assessment methods to evaluate these models and to assess and compare their out-of-sample performance. This has led to an overwhelmingly large set of volatility forecast models that a practitioner has to choose from, which is often an onerous task, given the degree of similarity between the models and the lack of consistent forecast selection rules. This paper seeks to address this issue by developing a validation framework to evaluate models based on the forecasts that they produce. Our framework is founded on the grounds of multiple comparison procedures within the model selection literature, and can easily be adapted to forecast evaluation problems.

1.1 Relevance of Model Selection to Volatility Forecast Problems

An often challenging task during the course of an econometric analysis is the comparison of different models for modeling the particular data set at hand. Depending on the specific application, the comparison can be based on different metrics. A common metric for comparing different models, as in criterion based model selection, is the likelihood score, often penalized with the number of parameters being estimated in some way, with the penalty being higher for models with a larger number of unknown parameters. Penalizing models based on the number of parameters is rooted in the idea that by using estimates rather than the true values of the parameters, the model will drift farther apart from reality. Therefore, the information criteria generally favor more parsimonious models with fewer parameters over more complex models. The validity of the information criteria as reliable model assessment tools, however, relies heavily on the assumption that the models are not misspecified, at least not intentionally. This is a rather strong assumption that is often violated, for example in the case of Quasi-Maximum Likelihood Estimation, where the likelihood function is misspecified. See Bollerslev, Engle, and Nelson (1994) and Sin and White (1996).

An alternative for comparing forecast models is the loss-evaluation of out-of-sample forecasts. This procedure requires the comparison of an appropriately defined loss function on the out-of-sample forecasts produced by each model, which is obtained by employing either a rolling or an expanding-window forecasting scheme. The choice between the rolling-window and the expanding-window forecasting schemes is often based on the assumptions surrounding the models and the data. For models with time-varying parameters, and for time-series data with rather short memories, the rolling-window scheme is deemed more appropriate in that it adapts to the most recent data while leaving out observations from the distant past that could become irrelevant as time moves forward. Moreover, by updating the parameter estimates upon each rolling forward of the training sample within the data set, the time-variability of the parameters can be appropriately accounted for. Once the series of forecast losses for each model is obtained, they can be compared to determine the model with the minimum overall loss.

One fundamental flaw with the direct comparison of the overall losses, however, is that it is prone to data snooping (or data mining). The effects of data snooping become prominent when multiple models are evaluated on the same data set, where some yield exceptionally smaller losses than others by chance alone, even if in reality they are equally good or even inferior. It is important, therefore, to control for the effects of data snooping in a hypothesis testing framework. The first to setup a testing framework of this kind was White (2000). The technique that White (2000) employs involves the pairwise comparison of all the models with a given benchmark, and then to determine whether the benchmark is outperformed by any of the other models in the sample in a multiple testing framework. The size of this test, which is known as the Family

Wise Error rate (FWE) is the main instrument by which tests of this kind control for data snooping.[1]

In addition to controlling for data snooping via the FWE, a desirable property for the statistical testing framework that we seek to develop is power. There are several different notions of power in statistical hypothesis testing. Perhaps the most familiar is the uniformly most powerful test, which has the greatest power among all tests of a given size. Even though the uniformly most powerful test can rather easily be achieved when testing for a single null hypothesis, when conducting a test of multiple hypotheses, power is restricted by the level enforced for the family wise error rate. For example, in the Bonferroni method for testing two hypotheses in a single multiple testing framework, each individual hypothesis can be conducted at most at a size $\alpha/2$ to ensure that the FWE is controlled at level α.[2] Nevertheless, some design considerations regarding the tests can lead to increased power while controlling the FWE at the same level. Two such mechanisms that we consider in this paper are the Step Wise Multiple testing procedure of Romano and Wolf (2005) (abbreviated as StepM) and the Model Confidence Set of Hansen, Lunde, and Nason (2011) (abbreviated as MCS).[3] Each of these testing mechanisms, which fall under the category of multiple comparison procedures, exploit the dependency structure of the test statistics involved to improve the power of the tests, while maintaining control over the FWE. These procedures will be discussed in detail in section 3.

A final consideration about power arises from the fact that most testing mechanisms rely on asymptotic results either for conducting the test itself or for imposing the regularity assumptions required to establish the validity of the tests. The scope of the asymptotic theory, however, is fairly limited in finite samples, and many of its results may not hold. This can significantly erode the power of the tests, unless a more powerful mechanism is implemented to estimate the distributions of the test statistics in finite samples. This can be achieved via an appropriately implemented bootstrap algorithm. As we will later show in section 3.5, the bootstrap can be used to directly estimate the density of the test statistics and to obtain the critical values for conducting the tests. We will use the bootstrap to this end, and to ultimately increase the power of our tests.

The remainder of this paper is organized as follows. Section 2 introduces the GARCH class of models and discusses their maximum likelihood estimation. Section 3 presents the validation framework and in particular the loss evaluation of forecast models. Stepwise Multiple Testing and the Model Confidence Set along with their bootstrap implementation are also covered in this section. Section 4 applies our framework to real market data and compares different volatility models for equities and FX rates. Later in the section, some diagnostic tests are presented for *ex-post* analysis of the results. Section 5 concludes and appendix A contains practical guidelines for code implementation.

2 Models of Conditional Heteroskedasticity

Early in the history of financial econometrics, it was discovered that financial return series, as statistical time-series objects exhibit certain peculiar patterns. These recorded patterns are collectively known as empirical regularities of the financial markets, and have served as the main driver of research in empirical finance. Many of the proposed models for modeling financial time-series and their dynamics are aimed at reproducing some or many of the empirical regularities that are commonly observed in the financial markets, and new discoveries on the front of empirical observation have helped refine and reassess already proposed models. One of the most prominent features concerning the empirical regularities of financial returns revolve around volatility. It is a well-established fact that financial returns exhibit volatility clustering with periods of high volatility followed by periods of high volatility and vice versa for periods of low volatility. Incidentally, in an effort to provide further refinements to the estimated variances of forecasts of macroeconomic aggregates, Engle (1982) discovers a conditionally heteroskedastic stochastic process that is capable of reproducing the volatility clustering effect that is so commonplace in financial markets. Less than four years later, in another

[1]The FWE is formally defined as the probability of incorrectly identifying at least one forecast model as superior to the benchmark (which again can happen due to the effects of data snooping).

[2]See Romano and Wolf (2005) section 2.3 and Romano and Wolf (2005) appendix D for illuminating examples.

[3]There are of course other testing procedures which have the same objectives over power and the FWE. Some examples include the SPA test of Hansen (2005), and the BRC of White (2000). Although a complete survey of all the existing methods is not admissible, the methods that we introduce in this paper are flexible and powerful enough to accommodate for most forecast selection problems that a practitioner faces on a daily basis.

ground-breaking paper, Bollerslev (1986) generalizes the ARCH model to the GARCH class (Generalized-Auto Regressive Conditional Heteroskedasticity), and lays the foundations for much of modern empirical finance, including conditional volatility forecasting in financial markets.

In the most general and abstract terms, an autoregressive conditionally heteroskedastic stochastic process $\{r_t\}_{t=1}^T$ is defined as:

$$r_t = h_t(\mathcal{F}_{t-1}; \theta) z_t, \tag{1}$$

where $\{z_t\}_{t=1}^T$ is an i.i.d. process stochastically independent from the process $\{r_t\}$, and with zero conditional mean $\mathbb{E}[z_t|\mathcal{F}_{t-1}] = 0$. The function $h_t(\mathcal{F}_{t-1}; \theta)$ is an $\mathcal{F}_{t-1}$-measurable function, parametrized by the parameter vector θ, with $\mathcal{F}_t$ denoting the σ-algebra induced by the process up to time t. The novelty in this specification of the return series is its non-linearity which accommodates for the correlation of non-linear transformations of the return series over time. This is plausible, since it is a very well-known fact that return series, although linearly uncorrelated, exhibit strong auto-correlation in absolute value or in quadratic terms. To see the non-linearity of the process in (1), consider the Hilbert space $\mathcal{H}_t^r$ representing the return process $\{r_t\}$ which is defined over a complete probability space $L(\Omega, \mathcal{F}_t, \mathcal{P})$, and let $\mathcal{H}_{t-1}^r \subset \mathcal{H}_t^r$ denote the Hilbert space spanned by the sub-process $\{r_s|s < t\}$. If we let further $\mathcal{H}_t^z$ denote the Hilbert space representing the i.i.d. process $\{z_t\}$, we can decompose $\mathcal{H}_t^r$ as:

$$\mathcal{H}_t^r = \mathcal{H}_{t-1}^r \otimes \mathcal{H}_t^z, \tag{2}$$

which defines the process $\{r_t\}$ as the tensor product between the sub-process $\{r_s|s < t\}$ and the i.i.d. process $\{z_t\}$, which is trivially non-linear.[4] The conditional variance of the return process is thus given by:

$$Var\,(r_t|\mathcal{F}_{t-1}) = h_t^2(\mathcal{F}_{t-1}; \theta), \tag{3}$$

which is heteroskedastic as it varies deterministically with time. The fundamental problem then becomes to specify a process for $h_t(\mathcal{F}_{t-1}; \theta)$ and to estimate the vector of parameters θ, which is the subject of the next sub sections. Finally, it is worth mentioning that different models differ in their specification of the function $h_t(\mathcal{F}_{t-1}; \theta)$ but their modeling logic remains fundamentally the same.

2.1 Canonical GARCH Class of Models

A standard specification for the conditional variance function in (3) is the GARCH(p,q) process which is given by:

$$h_t^2 = \omega + \sum_{i=1}^{q} \alpha_i r_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i}^2. \tag{4}$$

The GARCH process is extensively studied in the literature and its properties are well known. Key features of this process are its covariance-stationarity, imposed by the constraint $\sum_{i=1}^{q} \alpha_i + \sum_{i=1}^{p} \beta_i < 1$ and its symmetry, implying that both positive and negative returns influence future volatility equally. The baseline GARCH process has evolved over time and much of this evolution has been guided by empirical findings. A prominent finding that dates back to very early in the history of the literature is that negative returns have a larger impact on future volatility than positive returns of the same magnitude. This effect, commonly known as the leverage effect was first discovered by Black (1976) and later illustrated by Pagan and Schwert (1990) and Engle and Ng (1993) in the context of the news impact curve. A natural way to incorporate the leverage effect into the GARCH process is to adjust the coefficients of the lagged squared returns so that the coefficients are smaller when the returns are positive. This is the GJR-GARCH process of Glosten, Jagannathan, and Runkle (1993):

$$h_t^2 = \omega + \sum_{i=1}^{q} \left(\alpha_i + \gamma_i \mathbb{I}_{[r_{t-i}>0]}\right) r_{t-i}^2 + \sum_{i=1}^{p} h_{t-i}^2. \tag{5}$$

[4] It is worth mentioning that linear processes can always be decomposed as the direct sum of two or more orthogonal processes of the form: $\mathcal{H}_t^r = \mathcal{H}_{t-1}^r \oplus \mathcal{H}_{t-1}^{r\perp}$ where $\mathcal{H}_{t-1}^{r\perp}$ is the orthogonal complement of $\mathcal{H}_{t-1}^r$.

An alternative approach is to model the natural logarithm of the variance as in the EGARCH model of Nelson (1991):

$$log(h_t^2) = \omega + \sum_{i=1}^{q} \left[\alpha_i z_{t-i} + \gamma_i \left(|z_{t-i}| - \mathbb{E}\left[z_{t-i}\right]\right)\right] + \sum_{i=1}^{p} \beta_i h_{t-i}^2, \tag{6}$$

where $z_{t-i} = r_{t-i}/h_{t-i}$ is the standardized return. The leverage effect in the EGARCH process is manifested by the coefficient $\gamma_i < 0$. Another significant discovery was that absolute returns have a stronger serial correlation than squared returns. This was formally shown by Ding, Granger, and Engle (1993), and led to the proposal of the Asymmetric Power ARCH (APARCH) model:

$$h_t^\delta = \omega + \sum_{i=1}^{q} \alpha_i \left[|r_{t-i} - \gamma_i r_{t-i}|\right]^\delta + \sum_{i=1}^{p} \beta_i h_{t-i}^\delta. \tag{7}$$

By allowing the degree of the process, δ, to be a free parameter, the APARCH model is more flexible in capturing the dynamics of the conditional variance than other specifications. There exist many other models for the conditional variance, and the list of the models presented here is by no means exhaustive. Nevertheless, the models mentioned so far are among the most commonly used and nest many of the other specifications. See Ding, Granger, and Engle (1993) and Hentschel (1995).

2.2 Maximum Likelihood Estimation

The parameters of the GARCH-type processes[5] are naturally estimated via maximum likelihood. Under Fisher regularity conditions, the maximum likelihood estimator achieves the Cramer-Rao bound, and is therefore asymptotically efficient. An exception, however, may arise if the Fisher information matrix is singular, which happens if the fourth-order moment of the underlying process is infinite. In such cases, the (quasi)-maximum likelihood estimator may still be consistent, however, it will no longer be efficient. Under these circumstances, practitioners may prefer to use a heavy-tailed distribution for the likelihood function such as the Student-t. It must be noted, however, that this can introduce bias into the estimator. In fact, Hall and Yao (2003) show that within the class of maximum likelihood estimators, the Gaussian QMLE is the only distribution-free, and therefore consistent estimator. Any other misspecified likelihood function is bound to be asymptotically biased. Therefore, any possible gains in efficiency must be weighed against the potential bias due to the misspecification of the likelihood function.

Maximum likelihood estimation of the GARCH-type processes is a standard practice and can be found in most modern text books and related articles. See Hamilton (1994), Engle (1982), and Bollerslev (1986) among others. An important contribution is due to Hall and Yao (2003) where they show that the asymptotic distribution of the Gaussian QMLE is Levy-stable (although not Gaussian) and that its rate of convergence is $T^{1-\alpha/2}$, where α is the tail index of the density of standardized returns. Moreover, they show that the (robust) asymptotic covariance matrix of the quasi-maximum likelihood estimator is given by:

$$\Sigma = H^{-1} V H^{-1}, \tag{8}$$

where V is the asymptotic covariance matrix of the score function and H^{-1} is the inverse of the Hessian of the likelihood function.

Regardless of the choice of the estimator for GARCH-type processes, the conditions for the existence of the fourth-order moment of the underlying processes can be driven analytically. These conditions, however, are generally unknown and must be driven for each specific process individually.[6] Therefore, in practical applications, it is more convenient to employ an empirical method for testing for the finiteness of the fourth-order moment of the process *ex-post*, such as estimating the tail index of the standardized residuals. See section 4.2 for more details.

[5]We refer to the GARCH class of processes which include the GARCH process and its different varieties collectively as the GARCH-type processes.

[6]The necessary and sufficient condition for the GARCH(1,1) process for instance is: $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$.

3 Validation Framework

3.1 Rolling Window Estimation Scheme

A central difficulty in volatility forecasting is that the parameters of financial time-series models are rarely stable over long periods. Financial markets are frequently subject to structural breaks, changing persistence patterns, and volatility regime shifts. Consequently, parameter estimates obtained from distant historical observations may become uninformative or even detrimental for future forecasts. This issue is particularly important in risk management applications, where forecast accuracy depends critically on the model's ability to adapt to new market conditions. To address this problem, we employ a rolling window estimation scheme.

Let $\{r_t\}_{t=1}^{T}$ denote a sequence of demeaned returns and let W denote the size of the estimation window. For a given forecast model indexed by $m \in \mathcal{M}$, the model is estimated repeatedly on rolling subsamples of fixed length W. At each iteration t, the model is estimated using the observations:

$$\{r_{t-W+1}, r_{t-W+2}, \ldots, r_t\}, \tag{9}$$

and a one-step-ahead forecast of the conditional variance is produced:

$$\hat{h}^2_{t+1|t,m}. \tag{10}$$

The window is then shifted forward by one observation and the procedure is repeated recursively until the end of the sample is reached. Therefore, for each competing model, a sequence of out-of-sample volatility forecasts is generated:

$$\left\{\hat{h}^2_{W+1|W,m}, \hat{h}^2_{W+2|W+1,m}, \ldots, \hat{h}^2_{T|T-1,m}\right\}. \tag{11}$$

A further justification for the rolling window estimation scheme arises from the asymptotic theory of forecast comparison procedures. Let

$$d_{ij,t} = L_{i,t} - L_{j,t}, \tag{12}$$

denote the loss differential between two competing forecast models i and j, where $L_{i,t}$ and $L_{j,t}$ are the corresponding forecast losses at time t. Forecast comparison procedures such as the Superior Predictive Ability and the Model Confidence Set rely on the assumption that the sequence of loss differentials satisfies suitable stationarity and weak dependence conditions.

In practice, recursive parameter estimation on expanding samples may lead to substantial parameter uncertainty and unstable forecast loss dynamics, particularly in the presence of structural breaks. Rolling window estimation alleviates some of these issues by stabilizing the distribution of the forecast loss differentials and reducing the influence of distant observations. As emphasized by Giacomini and White (2006) and Hansen, Lunde, and Nason (2011), the estimation scheme itself forms an integral part of the forecast evaluation framework and must therefore be explicitly accounted for in predictive ability analysis.

3.2 Loss-Evaluation of Forecast Models

We evaluate and compare different forecast models by their average out-of-sample forecast losses. To this end, we would need to compute the loss entailed by each model by comparing the model forecasts with the observed volatilities. An immediate problem that arises, however, is that volatility is a latent variable and it cannot be observed. Therefore, we have to rely on (imperfect) proxies for the loss-evaluation of the forecast models. Further exacerbating the problem is the fact that loss functions are generally not robust to the noise involved in the proxies. This can lead to inconsistent evaluation schemes with often contradictory results, making the entire loss-evaluation procedure inefficient and in the most extreme case, altogether invalid. See Pagan and Schwert (1990), Bollerslev and Ghysels (1996) and Andersen, Bollerslev, and Lange (1999) for a discourse on this issue.

The inability to obtain consistent rankings of volatility forecast models has been looming over the gigantic volatility forecasting literature until A. J. Patton (2011), in his thorough analysis of loss-evaluation of volatility forecast models showed that the issue is related to the robustness (or the lack of) of the loss functions commonly employed for the evaluation of volatility forecast models, to the noise involved in the volatility proxies such as squared returns or realized volatility. In the absence of robustness, a loss-function evaluated on a series of imperfect proxies, yields as the optimal forecast a quantity other than the conditional expectation of the variable of interest. In other words, the noise involved in the proxies can distort and change the argmin of the loss function, making it incapable of measuring the distance between the model forecasts and the conditional expectations of the latent variances, even asymptotically.

A. J. Patton (2011) defines robustness as the property whereby the ranking of two forecasts obtained from imperfect volatility proxies is the same as the ranking that would be obtained from the infeasible latent variances. A robust loss function allows for the consistent comparison of forecast models, even if the latent variable of interest is not observable. It should be noted, however, that the use of robust loss functions does not automatically render the choice of the volatility proxy irrelevant. Robustness is a measure to guarantee that the comparison between forecasts is not systematically distorted by the use of an imperfect variable in place of the latent conditional variance. Nevertheless, the proxy still influences the noise of the evaluation, the precision of the loss differential estimations, and in the end, the following statistical inference. In empirical applications, the most widely used volatility proxies are squared returns and realized volatility. The former is simpler to compute and daily data are sufficient to provide availability, while the latter provides a more informative ex-post measure, when intraday data are available.

As mentioned earlier, a loss function $L(\cdot, \cdot)$ is defined as robust if the expected ranking between the two forecasts is preserved when the latent conditional variance is substituted by its observable proxy. A. J. Patton (2011) finds that among the most commonly used loss functions in the literature, only two satisfy this property: the Mean Squared Error (MSE) and the quasi likelihood loss implied by the Gaussian likelihood function (QLIKE). The MSE and the QLIKE are defined as follows. Formally, let $\hat{\sigma}_t^2$ be an ex-post observable proxy, and let $\hat{h}_{i,t}$ denote the conditional variance forecast produced by model $i \in \mathcal{M}$. Then, the Mean Squared Error is defined as:

$$L_{MSE}(\hat{\sigma}_t^2, \hat{h}_{i,t}^2) = (\hat{\sigma}_t^2 - \hat{h}_{i,t}^2)^2. \tag{13}$$

The MSE measures the quadratic distance between the variance proxy and the model forecast. Its main upside is the fact that it is immediately interpreted: wider forecasting errors are increasingly penalized, yielding an easy-to-compute and report metric. On the other hand, it is very sensitive to tail observations, since a few extreme volatility episodes can have a very high impact on the average loss.

The Quasi-Likelihood loss function, instead, is defined as

$$L_{QLIKE}(\hat{\sigma}_t^2, \hat{h}_{i,t}^2) = \log(\hat{h}_{i,t}^2) + \frac{\hat{\sigma}_t^2}{\hat{h}_{i,t}^2} \tag{14}$$

QLIKE has a natural quasi-likelihood interpretation, and it is particularly relevant since it penalizes forecasts that underestimate volatility, especially during high turmoil periods.

The joint application of MSE and QLIKE allows evaluating models' relative performance from complementary perspectives. MSE emphasizes absolute forecast error and is easy to interpret, while QLIKE is more linked to the logic of conditional variance models. If a model is competitive under both metrics, then the obtained ranking is more robust. If, instead, the results differ, this difference is still informative since it signals that the relative performance depends on the adopted accuracy definition. In the context of this paper, the main reference for the evaluation of the different models' performance will be these two loss functions.

From a practical standpoint, once the loss function is chosen, the comparison between two models i and j is performed through the loss differential:

$$d_{ij,t} = L(\hat{\sigma}_t^2, \hat{h}_{i,t}^2) - L(\hat{\sigma}_t^2, \hat{h}_{j,t}^2). \tag{15}$$

Hence, the loss differential series represents the central empirical object on which multiple comparison procedures are based. We further impose the following assumption on the loss differentials, which is strong enough to permit the employment of bootstrap methods to implement our forecast-validation framework.

Assumption 1: For some $r > 2$ and $\gamma > 0$, we have that $\mathbb{E}|d_{ij,t}|^{r+\gamma} < \infty$ for all $i, j \in M$ and that $\{d_{ij,t}\}$ is strictly stationary and α-mixing of order $-r/(r-2)$.

It should be noted that this assumption places restrictions on the loss differentials rather than directly on the individual loss series. Consequently, certain forms of structural instability and non-stationarity may still be admissible, provided that they affect competing models in a sufficiently similar manner so that the stationarity of the loss differential processes is preserved.

3.3 Stepwise Multiple Testing

In empirical forecast evaluation, competing models are often compared across a large number of alternative specifications. In such settings, superior forecasting performance may arise purely due to random variation rather than genuine predictive ability. This issue is commonly referred to as data snooping or data mining and represents one of the central challenges in empirical forecast evaluation.

Suppose that a collection of competing volatility forecast models

$$\mathcal{M} = \{\mathcal{M}_1, \mathcal{M}_2, \ldots, \mathcal{M}_m\} \tag{16}$$

is evaluated on the basis of their out-of-sample forecast losses. Even if all competing models possess identical predictive ability, repeated comparisons across many specifications increase the probability of falsely identifying at least one model as superior. Consequently, conventional pairwise testing procedures may produce misleading inference when applied repeatedly over a large universe of competing models.

The data snooping problem is particularly severe in empirical finance, where forecasting models are frequently modified, re-estimated, and compared across alternative specifications. As emphasized by Romano and Wolf (2005), repeated hypothesis testing without appropriate correction can substantially inflate the probability of false discoveries and lead to overoptimistic conclusions regarding predictive performance.

To address this issue, Romano and Wolf (2005) propose a stepwise multiple testing procedure that evaluates the predictive performance of the entire collection of models simultaneously while asymptotically controlling the familywise error rate.

To formalize the multiple testing problem, let

$$H_j : \mu_j \leq 0, \qquad H_j' : \mu_j > 0, \tag{17}$$

for $j = 1, \ldots, m$, denote the collection of hypotheses associated with the competing forecasting models. Here, μ_j represents the difference between the average forecast loss of a predefined benchmark or reference model and a model $j \in \mathcal{M}$. Therefore, under each individual null hypothesis, the benchmark yields an average forecast loss that is less than or equal to that of model $j \in \mathcal{M}$ against which it is being compared:

$$\mu_j = \mathbb{E}[d_{sj,t}] = \mathbb{E}[L_{s,t}] - \mathbb{E}[L_{j,t}] \leq 0, \tag{18}$$

where the index s corresponds to the benchmark or the reference model. The objective is to identify as many outperforming models as possible for which $\mu_j > 0$ while controlling the familywise error rate asymptotically.

Let

$$w_{T,j} \tag{19}$$

denote the test statistic associated with hypothesis H_j. The stepwise procedure begins by ordering the statistics from largest to smallest,

$$w_{T,r_1} \geq w_{T,r_2} \geq \cdots \geq w_{T,r_m}, \tag{20}$$

where the relabeling indices $(r_1, \ldots, r_m)$ correspond to the ordered hypotheses.

The central idea of Romano and Wolf (2005) is to construct joint confidence regions for the vector

$$(\mu_{r_1}, \ldots, \mu_{r_m})', \tag{21}$$

with nominal joint coverage probability $1 - \alpha$. In the first step, the joint confidence region takes the rectangular form

$$[w_{T,r_1} - c_1, \infty) \times \cdots \times [w_{T,r_m} - c_1, \infty), \tag{22}$$

where the common critical value c_1 is chosen such that the confidence region attains the required asymptotic coverage probability.

The hypotheses are then tested within this confidence region. Specifically, if the interval

$$[w_{T,r_j} - c_1, \infty) \tag{23}$$

does not contain zero, the corresponding null hypothesis H_{r_j} is rejected.

Unlike conventional single-step procedures, however, the stepwise methodology continues after the first rejection stage. Suppose that in the first step, the first R_1 hypotheses are rejected. Once the rejected hypotheses are removed, a new joint confidence region is constructed over the reduced hypothesis space,

$$[w_{T,r_{R_1+1}} - c_2, \infty) \times \cdots \times [w_{T,r_m} - c_2, \infty), \tag{24}$$

where the updated critical value c_2 is re-calibrated over the remaining hypotheses to obtain nominal joint coverage probability of $1 - \alpha$. The procedure is then repeated sequentially until no additional hypotheses are rejected.

The recursive updating of the confidence regions constitutes the primary source of the power improvement of the procedure. Standard single-step methods construct critical values that protect against the worst-case multiple testing scenario over the entire hypothesis space and therefore tend to be excessively conservative. In contrast, the stepwise procedure progressively reduces the dimension of the hypothesis space and recomputes the associated confidence regions at each stage. This leads to less conservative critical values and substantially improves the probability of rejecting false null hypotheses while still asymptotically controlling the familywise error rate.

In practice, the critical values $c_1, c_2, \ldots$ depend on the unknown sampling distribution of the statistics and are therefore approximated through bootstrap resampling methods. By estimating the joint dependence structure of the statistics directly from the data, the bootstrap procedure further improves the power of the tests relative to traditional correction methods such as Bonferroni-type procedures. Details are deferred to section 3.5.

The exposition above follows the basic StepM framework of Romano and Wolf (2005) based on generic test statistics $w_{T,j}$. In empirical applications, however, studentized statistics are often preferable because the variability of the test statistics may differ substantially across competing models. The studentized statistic is defined as

$$z_{T,j} = \frac{w_{T,j}}{\hat{\sigma}_{T,j}}, \tag{25}$$

where $\hat{\sigma}_{T,j}$ denotes a consistent estimator[7] of the standard deviation of $w_{T,j}$. The hypotheses are then ordered according to the studentized statistics rather than the raw statistics:

$$z_{T,r_1} \geq z_{T,r_2} \geq \cdots \geq z_{T,r_m}. \tag{26}$$

[7] A popular choice is a kernel estimator.

The procedure is identical to the one introduced earlier for the raw test statistics, and at each step k, the rectangular confidence region is formed as:

$$[w_{T,r_{R_{k+1}}} - \hat{\sigma}_{T,1} d_k, \infty) \times \cdots \times [w_{T,r_m} - \hat{\sigma}_{T,m} d_k, \infty), \tag{27}$$

where the common critical value d_k is chosen such that the confidence region has a nominal coverage probability of $1 - \alpha$. The null hypothesis H_{r_j} is rejected if the corresponding studendized confidence interval excludes zero:

$$0 \notin \left[w_{T,r_j} - \hat{\sigma}_{T,r_j} d_k, \infty\right). \tag{28}$$

Thus, studentization changes the scale on which the hypotheses are ordered and tested, but it does not change the logic of the StepM procedure. The method still constructs joint confidence regions, eliminates rejected hypotheses, and recomputes the critical values over the reduced hypothesis space. Its main advantage is that it accounts for heteroskedasticity and scale differences across test statistics, which makes the comparisons more balanced in finite samples.

The Model Confidence Set procedure developed by Hansen, Lunde, and Nason (2011), presented in the following section, extends this sequential testing logic to identify the subset of models that can be regarded as statistically indistinguishable from the each other in terms of predictive ability at a given confidence level.

3.4 Model Confidence Set

In many forecast comparison problems, no natural benchmark is available against which all competing specifications can be tested. In this setting, a procedure that starts from a set of plausible models and narrows it down to the best-performing alternatives is more appropriate than forcing a benchmark-based comparison. Moreover, as Hansen, Lunde, and Nason (2011) highlight, in many empirical applications, data do not contain sufficient information to univocally identify a single dominant model. In these cases, forcing the selection of a unique winner is likely to produce excessively confident conclusions compared to the evidence that is effectively available.

The Model Confidence Set (MCS) is a multiple comparison procedure developed by Hansen, Lunde, and Nason (2011), and it approaches the problem of model selection from the perspective of random set theory. The fundamental premise of this procedure is that given a random set of models, when there is not enough data available to identify a single superior model that outperforms every other model within the set, steps can be taken to at least remove the obvious under-performers and to reduce the set of the original models to a smaller confidence set that contains models whose relative performance is statistically indistinguishable from one another at a given confidence level. How smaller can this set get is a direct consequence of the informativeness of the available data. Therefore, in essence, the MCS can be considered as the equivalent, within the problem of model selection, of a confidence interval for an estimated parameter: when the data are informative, the final set can include few models and even reduce to just one; when, instead, data are less informative, the set remains broader, reflecting the inability to cleanly distinguish between the competing alternatives.

Let $\mathcal{M}^0$ be the initial set of candidate models and, as seen in previous sections, let

$$d_{ij,t} = L_{i,t} - L_{j,t} \tag{29}$$

denote the loss differential between models i and j at time t. Hence, the expected relative performance between the two models is defined as[8]:

[8]In contrast to the StepM procedure, there is no benchmark in here, therefore the pairwise loss differentials are defined for all possible pairs within the set. As will be discussed later in section 3.5, this leads to the definition of the average loss induced by all models within the set as the reference value for constructing the test statistics.

$$\mu_{ij} = E[d_{ij,t}]. \tag{30}$$

If $\mu_{ij} < 0$, model i presents a lower expected loss than model j and results preferable with respect to the considered loss function. Naturally, we can define a set $\mathcal{M}^*$ such that:

$$\mathcal{M}^* = \left\{ i \in \mathcal{M}^0 : \mu_{ij} \leq 0 \qquad \forall j \in \mathcal{M}^0 \right\}. \tag{31}$$

Hence, the set $\mathcal{M}^*$ contains all the models whose expected loss is no greater than that of any other model in the initial set. It is important to observe that the set $\mathcal{M}^*$ can include more than one element: the procedure does not assume, nor requires, that a unique best model exists.

The MCS procedure seeks to estimate $\mathcal{M}^*$ through the sequential application of a of an equivalence test and an elimination rule. Starting from the initial set of models $\mathcal{M}^0$, the equivalence test is set up to test the hypothesis that all the expected pairwise differential losses of the models within the set are equal to zero[9]:

$$H_{0,\mathcal{M}^0} : \mu_{ij} = 0 \quad \text{for all } i, j \in \mathcal{M}^0. \tag{32}$$

In the event that the null hypothesis is rejected, the elimination rule will be invoked to eliminate the model whose average forecast loss is the largest among all the models within the set. A distinctive feature of the elimination rule is that it is coherent with the equivalence test in that it eliminates from the set the model that yields the most extreme test statistic for the constituent hypotheses of the composite null hypothesis $H_{0,\mathcal{M}^0}$. The procedure then continues on the reduced subset and on all the subsets after that until the null hypothesis is not rejected anymore. In the most extreme case, the final subset is a singleton that contains the model whose average forecast loss is the smallest, and for which the null hypothesis of the equivalence test is by construction always true with probability exactly equal to one:

$$H_{0,\mathcal{M}^m} : \mu_{mm} = \mathbb{E}[L_{m,t}] - \mathbb{E}[L_{m,t}] = 0, \qquad Pr(H_{0,\mathcal{M}^m}) = 1 \tag{33}$$

The set of the remaining models at the end of the procedure is the MCS estimate of the set $\mathcal{M}^*$ and is referred to as the model confidence set. Key features of the model confidence set, denoted $\widehat{\mathcal{M}}^*_{1-\alpha}$, are its asymptotic coverage probability: $\liminf_{n\to\infty} Pr(\mathcal{M}^* \subset \widehat{\mathcal{M}}^*_{1-\alpha}) \geq 1-\alpha$, and its limiting power that equals one: $\lim_{n\to\infty} Pr(i \in \widehat{\mathcal{M}}^*_{1-\alpha}) = 0$ for all $i \notin \mathcal{M}^*$. Moreover, by conducting the equivalence test at each step of the procedure at the same level α, the family wise error rate is trivially bounded by that same level.

The MCS procedure also conveniently yields a p-value for every model in the original set, which determines the statistical threshold at which the models can belong to the model confidence set. Just as in classical hypothesis testing, the MCS p-values, denoted $\hat{p}_i \forall i \in \mathcal{M}^0$, quantify the probability of committing a type-I error by excluding the corresponding model from the confidence set. The elimination rule, naturally eliminates all the models whose MCS p-values are less than the pre-defined threshold α.

The MCS p-values are computed in a step-down procedure - just as in the StepM method - starting from the first subset of $\mathcal{M}^0 = \mathcal{M}_1$[10] and conducting the corresponding equivalence test for the hypothesis $H_{0,\mathcal{M}_1}$, proceeding to the next subsets defined as $\mathcal{M}^0 = \mathcal{M}_1 \supset \mathcal{M}_2 \supset \cdots \supset \mathcal{M}_m$ and testing the corresponding sequence of null hypotheses $H_{0,\mathcal{M}_2}$, $H_{0,\mathcal{M}_3}$, $H_{0,\mathcal{M}_m}$. The MCS p-value for each model is then defined as the maximum p-value obtained for any composite hypothesis during the entire procedure while the corresponding model constituted one of the constituent hypotheses. That is to say, after the first null hypothesis $H_{0,\mathcal{M}_1}$ is tested, the MCS p-value for every model is set to the p-value obtained for this hypothesis. If the hypothesis is rejected, and thus a model eliminated, the MCS procedure advances to the next step, where then the hypothesis $H_{0,\mathcal{M}_2}$ is tested. Given that in the previous step the model with the most extreme test statistic is eliminated, as is necessitated by the coherency requirement of the elimination rule, the p-value for $H_{0,\mathcal{M}_2}$

[9] It should be noted that the tests conducted in here are two-sided, whereas in the StepM procedure the tests are one-sided.

[10] At first sight this might appear to be rather confusing, but essentially, for the sake of argument, the equivalence tests are conducted on the subsets of $\mathcal{M}^0$ and not on $\mathcal{M}^0$ itself. Trivially, however, one of the subsets of any set, is the set itself, which in here is denoted by $\mathcal{M}_1$ and is exactly equal to $\mathcal{M}^0$.

is necessarily greater than or equal to the p-value for $H_{0,\mathcal{M}_1}$. Therefore, the sequence of p-values obtained from the MCS procedure is monotonically non-decreasing. If the p-value obtained for the hypothesis $H_{0,\mathcal{M}_1}$ is different (essentially greater) than the p-value obtained in the previous step, the MCS p-value for all the remaining models is updated and set to the p-value obtained at this step. The procedure continues, until the final subset $\mathcal{M}_m$ is reached, for which the p-value of the corresponding hypothesis, and consequently the MCS p-value, is by construction equal to one.

For all the simplicity regarding the decision rule for constructing the model confidence set on the basis of the MCS p-values, the computation of the p-values themselves is more intricate and involves a quadratic-from test of some kind, which requires a consistent estimator for the covariance matrix of the test statistics, and a large enough sample to allow for asymptotic inference. Even if reliable estimates of the covariance matrix are feasible to obtain and the sample size is large enough to satisfy the standard regularity conditions for asymptotic inference, the limiting distributions of the test statistics are non-standard and depend on nuisance parameters that may or may not be feasible to estimate.[11] Therefore, it would be more convenient to estimate the finite-sample distributions of the test statistics using a consistent bootstrap method, rather than drawing on asymptotic theory. The computation of the p-values then follows trivially, and it is the subject of the next sub section.

3.5 Bootstrap Implementation

Unless the sample size is large enough relative to the number of models being considered for the test statistics to uniformly converge to their asymptotic distributions, it is not possible to obtain precise estimates of the covariance matrix of the test statistics for carrying out the tests. Furthermore, the test statistics considered here have non-standard asymptotic distributions that depend on nuisance parameters. Therefore, it is convenient to use a bootstrap implementation that that does not require an explicit estimate of the covariance matrix and that it can handle the issue of nuisance parameters implicitly. By appropriately employing a bootstrapping algorithm, we can compute the critical values for constructing the confidence regions in the StepM procedure and directly estimate the MCS p-values.

The objective is to estimate the finite-sample distributions of the test statistics via the bootstrap. Given the stochastic nature of the loss series under evaluation, the appropriate scheme would be a time-series bootstrap. Among popular choices are the moving blocks bootstrap, the circular blocks bootstrap and the stationary bootstrap. The moving blocks bootstrap displays certain edge effects (data points at the edges of the sample have a lower probability of appearing in a particular bootstrap sequence than data points in the middle of the series), and together with the circular blocks bootstrap lacks stationarity.[12] The circular blocks bootstrap and stationary bootstrap overcome the issue of edge effects by wrapping the data into a circular sequence and the stationary bootstrap further resolves the non-stationarity problem by making the choice of the block sizes for drawing the bootstrap resamples random at each iteration. In the following, we consider all three bootstrap methods and provide an algorithm for choosing the block sizes (or the expected block size for the case of the stationary bootstrap) for drawing the bootstrap resamples.

We proceed by demonstrating the computation of the critical values for constructing the confidence regions for testing the null hypothesis of the StepM procedure and the MCS p-values via the bootstrap. As we shall see, the principal value on which the computations are founded is the maximum of the individual test statistics in either procedure. We then present an algorithm for choosing the appropriate bootstrap block sizes.

3.5.1 Computation of Critical Values for The StepM Procedure

Given the probability mechanism P that generated the test statistics that produce the confidence region in (22), the critical value for testing the null hypothesis at level α is naturally computed by taking the $1-\alpha$ quantile of the sampling distribution of $\max_{1\leq j\leq m}(w_{T,r_j} - \mu_{r_j})$. By considering the maximum of the

[11] The issue of nuisance parameters is due to the possible correlations between the test statistics, the very same property which we are trying to exploit to improve the power of our multiple testing framework.

[12] Regardless of the original series being stationary, stationarity is lost in the bootstrap resamples when blocks of fixed size are pieced together.

individual test statistics, the joint confidence region in (22) is ensured to have a nominal coverage probability of at least $1-\alpha$.[13] Therefore, the common critical value for the construction of the joint confidence region is given by:

$$c_1 \equiv c_1(1-\alpha, P) = \inf\left\{x : Prob_p\left\{\max_{1\leq j\leq m}(w_{T,r_j}-\mu_j) \leq x\right\} \geq 1-\alpha\right\}. \tag{34}$$

The problem, however, is that the probability mechanism P is generally unknown which makes the computation of the ideal quantiles infeasible. As a workaround to this problem, we will use the bootstrap to estimate P, and subsequently the critical value in (34). First consider a generic bootstrap mechanism $\hat{P}_T$, with the corresponding true parameter vector denoted by $\mu^*_T = \mu(\hat{P}_T)$. Then the critical value c_1 in (34) is estimated by $\hat{c}_1$ via the bootstrap as follows:

Algorithm 1 Computation of the Critical Values for The StepM Procedure via The Bootstrap

1: Given the ordered test statistics, generate B bootstrap data matrices $X_T^{*,1}, \ldots, X_T^{*,B}$.
2: **for** each bootstrap data matrix $X_T^{*,b}$, where $1 \leq b \leq B$ **do**
3: Compute the individual test statistics $w^{*,b}_{T,1}, \ldots, w^{*,b}_{T,m}$.
4: **end for**
5: **for** $1 \leq b \leq B$ **do**
6: Compute $\max^{*,b}_{T,1} = \max_{1\leq j\leq m}(w^{*,b}_{T,r_j} - \mu^*_{T,r_j})$.
7: **end for**
8: Compute $\hat{c}_1$ as the $1-\alpha$ empirical quantile of the B values $\max^{*,1}_{T,1}, \ldots, \max^{*,B}_{T,1}$.

For all the subsequent steps of the procedure, the critical values are computed via the same fashion as outlined in algorithm 1. For the case of the studentized test statistics, the procedure is similar. In particular, we compute the common critical value at the kth step of the procedure for the joint confidence region (27) as:

$$d_k \equiv d_k(1-\alpha, P) = \inf\left\{x : Prob_p\left\{\max_{R_{k+1}\leq j\leq m}(w_{T,r_j}-\mu_j)/\hat{\sigma}_{T,r_j} \leq x\right\} \geq 1-\alpha\right\}. \tag{35}$$

The only difference is that for every test statistic w_{T,r_j}, we also need to estimate the standard deviation $\hat{\sigma}_{T,r_j}$ to obtain the studentized test statistic. The choice of the estimator for the standard deviations is not unique and depends on the specific application. For i.i.d. data, for example, one can rely on sample estimators, whereas for time-series data one has to choose a robust estimator instead. In addition, a combination of bootstrap and kernel smoothing methods can be used[14]. For more details see Romano and Wolf (2005), Andrews (1991b) and Andrews and Monahan (1992). The critical values in (35) are estimated similarly via algorithm 2, which is an extension of algorithm 1.

With the common critical values for the raw and studentized test statistics computed via algorithms 1 and 2, we construct the joint confidence regions for the kth step of the procedure as:

$$[w_{T,r_{R_{k+1}}} - \hat{c}_k, \infty) \times \cdots \times [w_{T,r_m} - \hat{c}_k, \infty), \tag{36}$$

for the raw test statistics and:

$$[w_{T,r_{R_{k+1}}} - \hat{\sigma}_{T,1}d_k, \infty) \times \cdots \times [w_{T,r_m} - \hat{\sigma}_{T,m}d_k, \infty), \tag{37}$$

for the studentized test statistics.

[13] Readers will realize that this is more restrictive than if we were to test each hypothesis individually, in which case for the other test statistics we would choose a smaller critical value. However, in that framework, we would need to conduct the individual tests at a nominal level strictly less than α in the lines of the Bonferroni method to ensure that the FWE is bounded by α.

[14] This is the method that we use for our empirical analysis. Details are panned out in the source files of Sheppard (2025).

Algorithm 2 Computation of the Studentized Critical Values for The StepM Procedure via The Bootstrap

Given the ordered test statistics, generate B bootstrap data matrices $X_T^{*,1}, \ldots, X_T^{*,B}$.
for each bootstrap data matrix $X_T^{*,b}$, where $1 \leq b \leq B$ **do**
    Compute the individual test statistics $w_{T,1}^{*,b}, \ldots, w_{T,m}^{*,b}$, and the corresponding standard errors $\hat{\sigma}_{T,1}^{*,b}, \ldots, \hat{\sigma}_{T,m}^{*,b}$.
end for
for $1 \leq b \leq B$ **do**
    Compute $\max_{T,k}^{*,b} = \max_{R_k \leq j \leq m}(w_{T,r_j}^{*,b} - \mu_{T,r_j}^{*})/\hat{\sigma}_{T,r_j}^{*,b}$.
end for
Compute $\hat{d}_k$ as the $1-\alpha$ empirical quantile of the B values $\max_{T,k}^{*,1}, \ldots, \max_{T,k}^{*,B}$.

3.5.2 Computation of MCS p-values

To compute the p-values associated with each model in the MCS procedure, we first need to develop a test for testing the null hypothesis $H_{0,\mathcal{M}}$ at each step of the process. Given the expected relative loss differentials $\mu_{ij} = \mathbb{E}[d_{ij,t}] = \mathbb{E}[L_{i,t} - L_{t,j}]$, let us define the relative sample loss statistics $\bar{d}_{ij} = T^{-1}\sum_{t=1}^{T} d_{ij,t}$ and $\bar{d}_i = m^{-1}\sum_{i\in\mathcal{M}} \bar{d}_i$. Here $\bar{d}_{ij}$ is the sample counter part of the expected loss of model i relative to model j and $\bar{d}_i$ is the sample loss of model i relative to the average across all models in $\mathcal{M}$. Recall from section 3.4 that under the null hypothesis $H_{0,\mathcal{M}}$, the expected relative loss differentials are all equal to zero:

$$H_{0,\mathcal{M}} : \mu_{ij} = 0 \quad \text{for all} \quad i, j \in \mathcal{M}. \tag{38}$$

The statement $\mu_{ij} = 0\ \forall i, j \in \mathcal{M}$ is equivalent to $\mu_i = 0\ \forall i \in \mathcal{M}$ where $\mu_i \equiv \mathbb{E}[\bar{d}_i]$. Therefore, the null hypothesis can be restated as:

$$H_{0,\mathcal{M}} : \mu_i = 0 \quad \text{for all} \quad i \in \mathcal{M}. \tag{39}$$

We can test the null hypothesis stated above by constructing the following test statistic for every model $i \in \mathcal{M}$:

$$t_i = \frac{\bar{d}_i}{\sqrt{\widehat{\text{var}}\left(\bar{d}_i\right)}}. \tag{40}$$

Since it is enough for just one of the relative expected loss differentials to be different from zero to reject the null hypothesis, we choose the test statistic with the largest value among all t_i and concentrate all the power in a single test. Therefore, to test the null hypothesis $H_{0,\mathcal{M}}$ at each step of the MCS procedure, we compute the test statistics t_i for every model and then choose $T_{\max,\mathcal{M}} = \max_{i\in\mathcal{M}} t_i$ to carry out the test.

The asymptotic distribution of the test statistics t_i is nonstandard and this can readily be seen from the way that the relative average sample losses d_i are constructed. Each relative sample loss is defined as the difference between the average sample loss of model i and the average sample loss across all models, meaning that both the average relative sample losses, and consequently the test statistics are correlated. Therefore, the (joint) asymptotic distribution of the test statistics depends on a nuisance correlation matrix that makes the inference problem nonstandard.[15] In this case, implementing an appropriate bootstrapping mechanism not only resolves the issue related to the nuisance correlation matrix, but also improves the efficiency of the tests by furnishing p-values for each test based on direct estimates of the finite sample distributions of the test statistics. As already mentioned in the previous section, any of the moving blocks bootstrap, circular bootstrap, or the stationary bootstrap methods can be used given an appropriately sized (expected) block size.

Once the p-value for the test statistic $T_{\max,\mathcal{M}}$ is estimated, the MCS procedure proceeds naturally. At each step of the procedure, if the null hypothesis is rejected, the MCS p-values are updated and the model

[15] It should be noted that the correlation structure of the test statistics per se is of no interest at all, and any attempt at estimating the correlation matrix would be an unfruitful squandering of information that will only result in losing degrees of freedom.

corresponding to $T_{\max,\mathcal{M}} = \arg\max_i t_i$ is eliminated from the set. If the null hypothesis is not rejected, the procedure terminates and the set of remaining models will be returned as the model confidence set.

3.5.3 Choice of Block Sizes

In order to implement a time series bootstrap, that is, one of the moving blocks, circular blocks, or the stationary bootstrap, it is required to choose an appropriate bootstrap block size b. The practical procedures for choosing the block size are mostly ad-hoc, under the light of the asymptotic requirements which involve $b \to \infty$ and $b/T \to 0$ as $T \to \infty$. We use Sheppard's (Sheppard (2025)) implementation of the algorithm proposed by Politis and White (2004) and A. Patton, Politis, and White (2009) to choose the optimal block size for the bootstrap. The algorithm involves a tuning parameter m, which is chosen as the first integer value where k_n consecutive autocorrelations of the input series are all inside a conservative band of $\pm 2\sqrt{log_{10}(T)/T}$ where T is the sample size. The maximum value of m is set to $\lceil \sqrt{T} + k_n \rceil$ where $k_n = \max[5, log_{10}(T)]$. Having chosen the tuning parameter m, the variance of the time series is estimated via the kernel estimator:

$$\hat{\sigma}^2 = \sum_{k=-m}^{m} h(k/m)\hat{\gamma}_k, \tag{41}$$

where $\hat{\gamma}_k$ is the sample estimator of the k^{th} autocovariance of the series which is given by:

$$\hat{\gamma}_k = \frac{1}{T} \sum_{j=k+1}^{T} (x_j - \bar{x})(x_{j-1} - \bar{x}). \tag{42}$$

The kernel function is given by $h(x) = \min(1, 2(1-|x|))$. Additionally, the autocovariance-generating function is estimated via the same kernel function as:

$$g = \sum_{k=-m}^{m} h(k/m)|k|\hat{\gamma}_k. \tag{43}$$

Finally, the optimal block size is computed as:

$$b_i^{OPT} = \left(\frac{2g^2}{c_i(\hat{\sigma}^2)^2} T \right)^{1/3}. \tag{44}$$

In the expression above, the constant c_i is set to 2 for the stationary bootstrap and 4/3 for the circular and the moving blocks bootstrap.

4 Empirical Analysis

4.1 Results

4.2 Diagnostic Tests

5 Conclusion

A Code Implementation

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